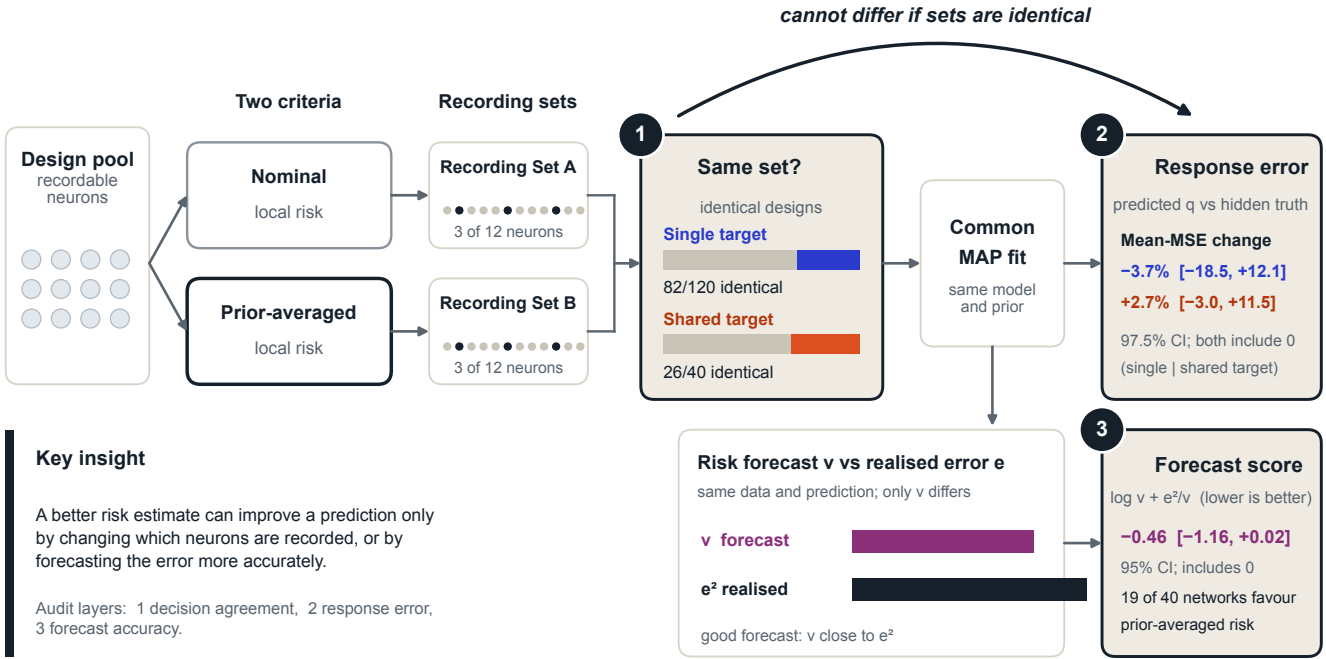

cannot differ if sets are identical
Two criteria
Recording sets
Design pool
recordable neurons
Nominal
local risk
Prior-averaged
local risk
Recording Set A
3 of 12 neurons
Recording Set B
3 of 12 neurons
1
Same set?
identical designs
Single target
82/120 identical
Shared target
26/40 identical
Common MAP fit
same model and prior
2
Response error
predicted q vs hidden truth
Mean-MSE change
−3.7% [−18.5, +12.1]
+2.7% [−3.0, +11.5]
97.5% CI; both include 0
(single | shared target)
Risk forecast v vs realised error e
same data and prediction; only v differs
v forecast
e² realised
good forecast: v close to e²
3
Forecast score
log v + e²/v (lower is better)
−0.46 [−1.16, +0.02]
95% CI; includes 0
19 of 40 networks favour prior-averaged risk
Key insight
A better risk estimate can improve a prediction only by changing which neurons are recorded, or by forecasting the error more accurately.
Audit layers: 1 decision agreement, 2 response error, 3 forecast accuracy.